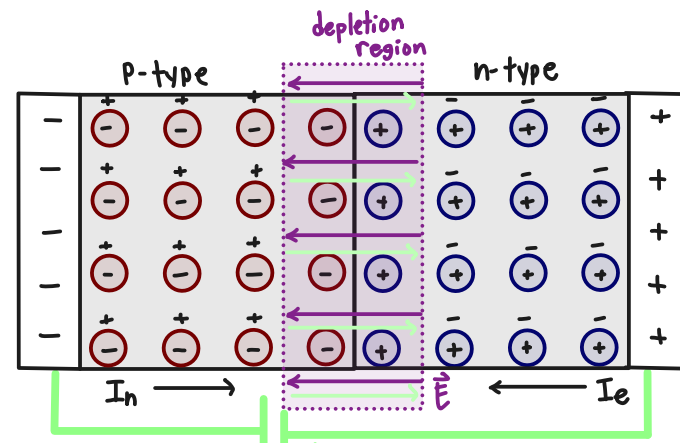


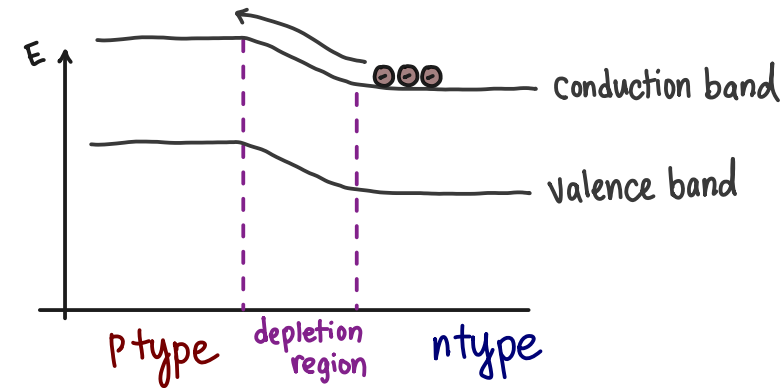
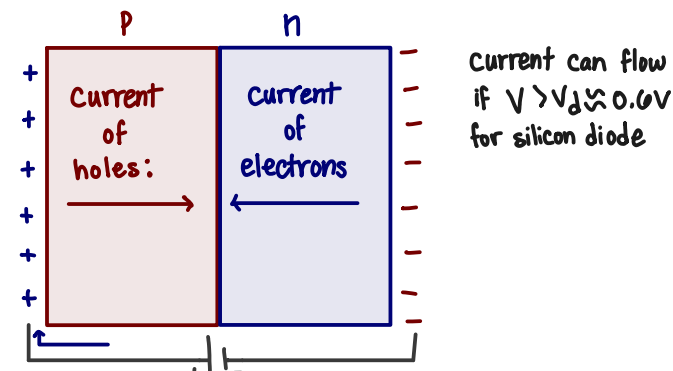
Review: Diode:



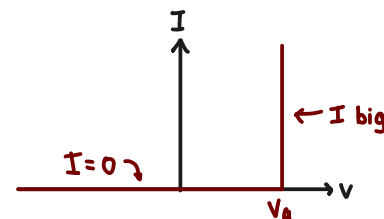
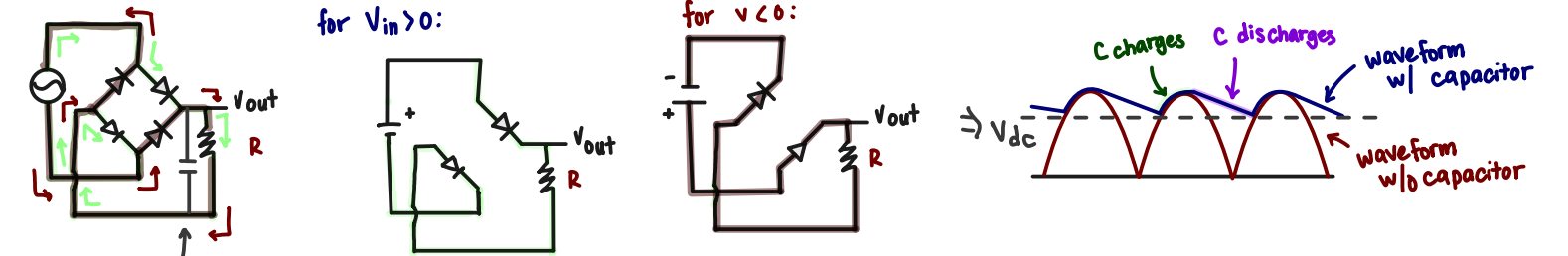
P-type dopants
• extra 'holes'

n-type dopants
extra electrons

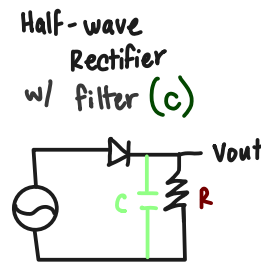
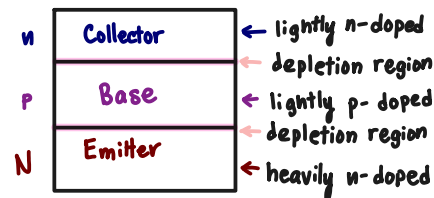
depletion region
- no extra carriers



Full Wave Rectifier



Bipolar Junction Transistor (BJT)
or npn transistor



when C is charging
→ OA to R (discharge path)

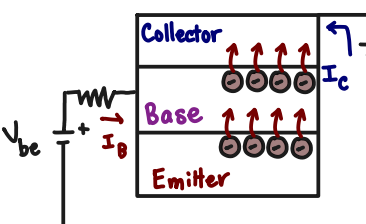
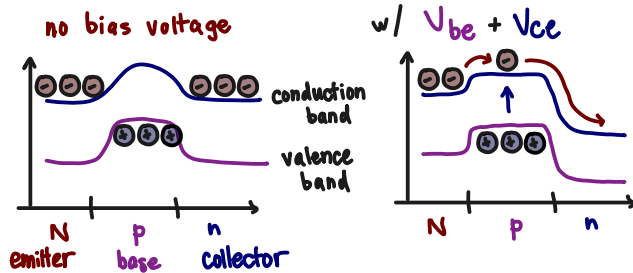
when discharging

$$V_{out} = (V_{in} - 0.6) e^{-t/RC} = V_{max} (1 - \frac{t}{RC})$$

for $R \gg T = \frac{2\pi}{\omega}$, $e^{-t/RC} \approx (1 - \frac{t}{RC})$

$\Delta V = \text{Ripple} = \text{peak-to-peak}$

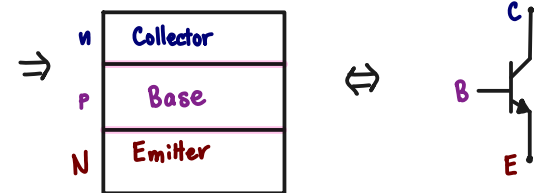
$$\Delta V = V_{max} \left(\frac{\Delta t}{RC} \right) = V_{max} \frac{2\pi}{\omega RC} = I_{max} R \frac{2\pi}{\omega RC} = \frac{I_{max}}{fC} \quad (f = \text{frequency})$$



* some e^- can recombine with holes in base creating I_B

* some e^- are pushed from B to C by applied V_{ce} and create I_C

Cross section of npn transistor
goal: $I_C \gg I_B$, $I_E = I_B + I_C \approx I_C$ (to make good amp)



$I_E = I_B + I_C$

$I_C \propto I_E$ $\alpha \approx 1$

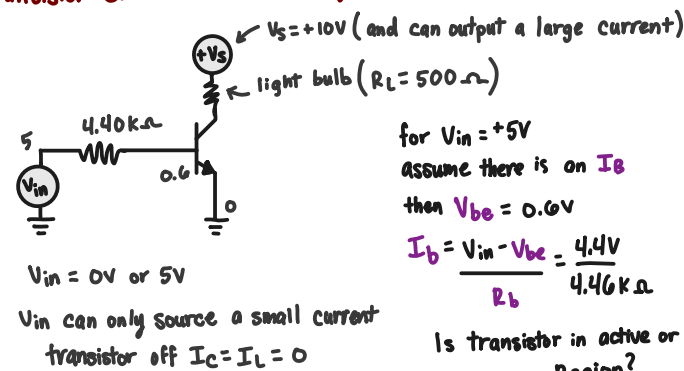
$I_B = (1 - \alpha) I_E$

$\frac{I_C}{I_B} = \frac{\alpha}{1 - \alpha} = \beta \gg 1$

$I_C = I_E = \beta I_B$

$\beta \approx 50-200$ for typical transistors in our lab

Transistor Switch Saturation Region

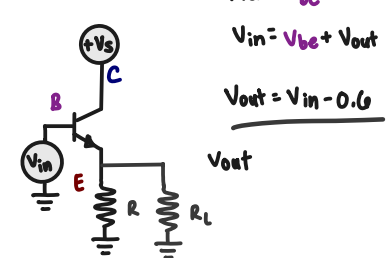


for $V_{in} = +5V$
assume there is an I_B
then $V_{be} = 0.6V$
 $I_B = \frac{V_{in} - V_{be}}{R_b} = \frac{4.4V}{4.4k\Omega}$

Is transistor in active or saturation Region?

then $I_C = 100mA \rightarrow \text{max Voltage across } R_L \approx 10V$
max Current $I_{max} = \frac{10V}{500\Omega} = 20mA$
 $\beta = 20$

Emitter Follower:



then $V_{be} = 0.6V$
 $V_{in} = V_{be} + V_{out}$
 $V_{out} = V_{in} - 0.6$

Assume $V_{in} > 0.5V$
and $\frac{V_S}{R} \gg \beta I_B$

→ active region: assume β

* what good is this?

V_{out} remains the same even if I add on output load

